

Prottus Manna

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prottus-visual-showcase.vercel.app | Kolkata, India | Open to Relocation | Fresher | Application Security Engineering Intern

Education

Institute of Engineering and Management (IEM), Kolkata – MAKAUT	2022 – 2026
<i>Bachelor of Technology (B.Tech), Information Technology/ CS (Graduate)</i>	<i>SGPA: 9.00 in Final Semester, Aggregate: 8.35</i>
National Gems Higher Secondary School, Behala	2022
<i>Class XII – ISC</i>	<i>Aggregate: 76%</i>
National Gems Higher Secondary School, Behala	2020
<i>Class X – ICSE (Indian Certificate of Secondary Education)</i>	<i>Aggregate: 86%</i>

Work Experience

- Fraud Guard AI – Zidio Development** | *Python, React.js, Node.js, Express.js, REST API* June 2025 – July 2025
- Designed and deployed a full-stack production application with REST API integrations to third-party email/data services, structuring the backend for modular, independently deployable services.
 - Built an **automated** retraining pipeline triggered by analyst feedback, enabling continuous deployment of model updates with zero manual intervention – improving detection accuracy by **~18%** and system **reliability**.
 - Applied Agile/Scrum practices and **CI/CD**-oriented delivery across the full SDLC in **collaboration** with a cross-functional team; authored technical design docs and API specs supporting **scalability**, maintainability, and long-term **operational excellence**.
- Vision AI Suite - Zidio Development** | *Python, PyTorch, CNN+LSTM, U-Net, OpenCV, Streamlit* May 2025 – June 2025
- Architected and deployed a full-stack concurrent multi-model deep learning pipeline running three specialised neural networks simultaneously: a CNN+LSTM encoder-decoder for image captioning, a U-Net for semantic segmentation, and a custom object detection module.
 - Deployed the complete AI application pipeline as a real-time interactive Streamlit app, supporting live video and image inputs with optimised low-latency concurrent execution across all three model architectures.
 - Created an efficient multi-model serving strategy to minimise GPU memory usage and reduce latency – demonstrating the ability to triage and resolve system performance issues at the hardware and service layer.

Key Projects

- Optimus AI — Agentic Voice Assistant** July 2025 – Present
- Python, LangChain, LangGraph, LLMs, RAG, edge-TTS, SQLite, Selenium, PyAutoGUI, Pydantic*
- Architected an end-to-end **agentic AI pipeline** mapping a single spoken command to multi-step action: LLM intent router → Pydantic-typed tool registry (20+ tools) → supervised **self-healing executor** (LLM-guided retry on failure) → natural-language response generator, with wake-word detection and voice barge-in.
 - Built a **pluggable multi-provider LLM chain** (NVIDIA NIM → local Ollama) behind a uniform interface with automatic health-based fallback, and a **date-aware RAG pipeline** (LLM query reformulation, recency filters, multi-source web grounding) for hallucination-free real-time answers on live facts, prices, and news.
 - Engineered a low-latency **streaming TTS engine** (4-worker parallel synthesis + sentence chunking + instant “thinking” filler) and **runtime OS-level app discovery** fusing 5 sources (Start Menu shortcuts, registry App Paths, Uninstall entries, PATH, web fallback) with fuzzy reconciliation – zero hardcoded app lists, portable across Windows machines.
 - Implemented a **disk-first code-generation agent** (plan → 70B-model streaming generation → atomic save → IDE focus-guarded typing → syntax verification with LLM-assisted regeneration) and **SQLite-backed memory** for conversation continuity and preference learning across sessions.

Achievements & Recognition

- Achieved the rank of 283 among 17,040 participants in **Naukri Campus Techquest #32**
- Rewarded for being the solo engineer to build 3 production-grade application engineering systems at **Zidio Development**.
- Selected by IEM Kolkata to visit **National University of Singapore(NUS)**; received specialised training in **Artificial Intelligence, Machine Learning, IoT, and Data Analytics** at the **National University of Singapore(NUS)**.
- Coding Efficiencies with real-time projects: Solved **10,000+ DSA questions and software development** coding problems.

Technical Skills

Programming Languages: Python, Java, C/C++, HTML/CSS, Typescript, Javascript
Software Engineering: React.js, Node.js, Express.js, Docker, JSON, HTTP/HTTPS, Streamlit, Information Security
AI / ML & LLMs: PyTorch, TensorFlow, scikit-learn, Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, NLP, CNN, LSTM, U-Net, Transformer Models, Supervised/Unsupervised Learning, Automation, LLM Training & Retraining with RL
System & Application Integration: REST API, Third-Party API Integration, Sync Integration, Automated Retraining Pipelines
Databases: MongoDB (NoSQL), MySQL, PostgreSQL, SQL Query Optimisation Microsoft office full (Excel, Word, Powerpoint)
Cloud & DevOps: Microsoft Azure (Fundamentals), Git, GitHub, Linux, Amazon Web Services (AWS), Architecture
Core CS Concepts: Data Structures & Algorithms (DSA), OOP, System Design, SDLC, Operating Systems, DBMS, Computer Networks, Data Mining, Engineering, Anomaly Detection, Data Protection, Kubernetes, Auditing, Problem Solving Skills
Engineering Practices: Clean/Modular Code, Technical Documentation, Issue Triage & Debugging, Test Documentation, Code Review, Anomaly Detection, Customer Satisfaction, Customer Intelligence, CI/CD Pipelines, Agile/Scrum, Dashboards, Mentorship
Full-Stack Web Development Frontend/Backend, Hosting, Monitoring & Logging, Performance Optimization, Security, Testing, DevOps & Deployment, API Integration & Communication, Version Control & Collaboration, Critical Thinking, Leadership